



Schizophrenia and Related Psychotic Disorders

Heather I. Milliken, MDCM, FRCPC, CSPQ, FCPA

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Introduction

Psychoses are brain disorders that cause a distortion of, or loss of contact with, reality and affect a patient's ability to think, feel, perceive and act. Psychotic symptoms include the following:

- Positive symptoms such as delusions, hallucinations, disorganized thinking, speech and behaviour
- Negative symptoms such as social withdrawal, apathy, anhedonia and emotional blunting
- Cognitive symptoms such as impaired attention, concentration and memory
- Mood symptoms such as dysphoria, anxiety and emotional lability

Approximately 3% of people worldwide will experience at least 1 psychotic episode during their lifetime. See [Table 1](#) for a list of differential diagnoses. As symptoms can evolve during the early phase of illness, consider the initial diagnosis provisional, and re-evaluate as the disease progresses. The age of onset of schizophrenia spectrum psychotic disorders is usually late adolescence/early adulthood. There is increased emphasis on earlier detection and intervention^[1] because the duration of untreated psychosis can be associated with poorer outcomes in both the short and long term.^{[2][3]}

Table 1: Differential Diagnosis of Psychotic Episodes^[4]

Disorder	Characteristics
Schizophrenia	Signs of illness for ≥6 months; psychotic symptoms (delusions and/or hallucinations and/or disorganized speech) for ≥1 month; social/occupational dysfunction
Schizophreniform disorder	Similar to schizophrenia except duration of illness <6 months and ≥1 month
Schizoaffective disorder	Uninterrupted period of illness in which symptoms of schizophrenia and major mood episode occur concurrently; during lifetime duration of illness, ≥2 wk of delusions or hallucinations in absence of major mood episode; major mood episode is present >50% of total duration of illness
Delusional disorder	Nonbizarre delusions for ≥1 month and patient does not meet criteria for schizophrenia
Brief psychotic disorder	Psychotic symptoms for ≥1 day but <1 month; may or may not be related to marked stressor; eventual full return to premorbid level of functioning

Disorder	Characteristics
Substance-induced psychotic disorder	Delusions or hallucinations develop during or within 1 month of substance intoxication or withdrawal or are etiologically related to medication use and are not better accounted for by another psychotic disorder
Psychotic disorder associated with another medical condition	Delusions or hallucinations are a direct physiological consequence of a medical condition and occur in absence of delirium
Major depression or bipolar disorder with psychotic features	Delusions or hallucinations occur during a major depressive or manic episode
Psychotic disorder not elsewhere classified	Psychotic symptoms present but criteria for specific disorder not met or there is insufficient or contradictory information

Prevention

Increasingly, research has focused on the identification and treatment of individuals who are at “clinical high risk” of developing a psychotic disorder or may be in the prodromal phase.^{[5][6][7]}

Primary care providers can play an important role in early detection.^{[8][9]} A number of tools, including structured interviews and rating scales, have been developed to identify individuals who may meet the criteria for being at clinical high risk of progressing to a psychotic disorder.^[10] Examples include the [Prevention through Risk Identification, Management, and Education \(PRIME\) screening test](#); the Structured Interview for Prodromal Syndromes (SIPS); and the Scale of Prodromal Syndromes (SOPS).

Common prodromal psychosis symptoms (in order of decreasing frequency) include:^[11]

- Reduced concentration and attention
- Reduced drive and motivation; lack of energy
- Depressed mood
- Sleep disturbance
- Anxiety
- Social withdrawal
- Deterioration in functioning
- Irritability

The primary goal of treatment for individuals who meet the criteria for a prodromal psychosis syndrome is to prevent the progression to psychosis and to alleviate current and distressing symptoms, such as anxiety and depression. Canadian schizophrenia guidelines recommend a staged approach, beginning with cognitive behavioural therapy (CBT) and progressing to low-dose second-generation antipsychotics for those who do not respond to psychological interventions.^[12] For more information on treatment options, see [Therapeutic Choices](#).

If an individual is deemed to be at clinical high risk of developing a psychotic disorder or may be in the prodromal phase, referral to a specialized [early psychosis program](#) (if available), psychiatrist or community mental health team is recommended. The [Canadian Consortium for Early Intervention in Psychosis](#), a national and bilingual organization of clinicians who are involved with early psychosis programs across Canada, also provides access to various tools and resources.

Goals of Therapy

- Prevent progression to psychosis in patients at clinical high risk
- Reduce psychotic agitation in acute episodes
- Achieve remission of positive, negative, mood and cognitive symptoms
- Reduce risk of psychiatric comorbidity, particularly suicide and depression
- Reduce risk of physical comorbidity including metabolic syndrome and cardiovascular disease
- Reduce risk of substance abuse
- Reduce risk of harm to self and others
- Reduce risk of social isolation due to alienation from family, friends and social supports
- Facilitate recovery of functioning and healthy development
- Prevent or minimize recurrence of psychotic episodes

Investigations

- Family physicians are often the initial contact for patients experiencing a first psychotic episode. They should have a high index of suspicion in any young individual who is experiencing persistent changes in behaviour, mood and functioning, especially in the presence of other risk factors, such as:
 - substance abuse
 - family history of mental illness, particularly a history of psychotic disorders
 - childhood trauma
 - obstetrical complications at birth (e.g., extreme prematurity, hypoxia/ischemia)
- Individuals with suspected first-episode psychosis require urgent services. If available, refer to a psychiatrist, community mental-health clinic or specialized early psychosis program.^[13]
- In addition to the common prodromal symptoms previously discussed, signs and symptoms of possible first-episode psychosis include:^[11]
 - rapid fluctuations in mood (emotional lability) or showing very little emotion or facial expression
 - unreasonable suspiciousness
 - insomnia, restlessness and pacing at night
 - unusual or bizarre behaviour
 - unusual perceptual experiences including hypersensitivity, illusions and/or brief intermittent hallucinations
 - difficulties in thinking, such as organizing and/or expressing thoughts
- Substance use (particularly cannabis) is common in first-episode psychosis. Individuals may therefore be misdiagnosed with a substance-induced psychosis and not receive appropriate ongoing treatment. Cannabis use can trigger the onset of a schizophrenia spectrum disorder in genetically vulnerable individuals^[14] and in individuals with schizotypal disorder.^[15] See [Substance Abuse](#) for more information.
- A high index of suspicion for a functional psychosis (not substance-induced) is warranted, even in the presence of substance use, if:

- symptoms precede the onset of substance use
- symptoms are bizarre or there is marked thought disorder
- symptoms persist beyond the period of intoxication or withdrawal
- In an acute psychotic episode, assess the nature and extent of psychopathology:
 - thorough history of presenting problems with special attention to onset and course of prodromal symptoms
 - onset, characteristics and severity of psychotic symptoms
 - changes in behaviour and functioning
 - history of any suicidal ideation or behaviour and/or aggressive/violent ideation or behaviour
 - history of substance use/abuse in relation to onset and course of psychotic symptoms
- A thorough mental status examination is essential. Competency to consent to treatment needs to be assessed in all acutely psychotic individuals.
- Obtain information from as many sources as possible, since individuals with psychotic disorders are often poor historians. Interview family members whenever possible with the consent of the individual. If a patient is unwilling to give consent to interview family members, a clinician may still accept collateral information provided by a family member (e.g., in a phone call) but cannot divulge any information that would constitute a breach of patient confidentiality. If a patient is judged unable to give informed consent, then legally a substitute decision-maker (usually a family member) has to be contacted in order to provide informed consent on behalf of the patient.
- A variety of clinical rating scales can be used at baseline and repeated periodically to monitor for symptomatic and functional recovery following an acute psychotic episode:
 - The Clinical Global Impression Scales for Severity (CGI-S)^[16] and for Change (CGI-C)^[16] and the Social and Occupational Functioning Assessment Scale (SOFAS)^[17] for documenting changes over time are easy to use.
 - Training is required in order to reliably use the [Brief Psychiatric Rating Scale](#) (BPRS) and the Positive and Negative Syndrome Scale (PANSS). Alternatively, the semistructured interview guide SCI-PANSS^[18] takes approximately 30–40 minutes to complete and can be helpful in eliciting signs and symptoms of psychopathology.
- Patients with only partial symptomatic and/or functional recovery following an acute psychotic episode require diagnostic reassessment by a psychiatrist.
- Appropriate investigations, based on both the phase of the psychotic disorder and individual patient characteristics, are listed in [Table 2](#).

Table 2: *Investigations and Monitoring of Psychoses*^{[19][20]}

Parameter	Tools for Screening	Phase of Illness, Recommended Monitoring/Frequency
Psychopathology (positive/negative symptoms, risk of suicide/aggression)	CGI-S, CGI-C, BPRS, PANSS	First episode: baseline then weekly for first 4–8 wk; more often if clinically indicated Recurrent acute episode: baseline then weekly for first 4–8 wk; more often if clinically indicated Stabilization phase: monthly for first 6 months following first or recurrent acute episode; more often if clinically indicated Stable phase: every 3 months for individuals with good symptomatic and functional recovery and medication adherence; more often for individuals with poor medication adherence, residual symptoms, poor functioning, substance abuse

Parameter	Tools for Screening	Phase of Illness, Recommended Monitoring/Frequency
Substance use	CAGE, CAGE-AID, AUDIT	First episode: baseline Recurrent acute episode: baseline Stabilization phase: at every patient visit or as clinically indicated Stable phase: at every patient visit or as clinically indicated
Level of functioning (activities of daily living, social and occupational functioning)	SOFAS	First episode: premorbid level of functioning; baseline assessment of current functioning Recurrent acute episode: baseline assessment of current functioning Stabilization phase: monthly for first 6 months following first or acute episode Stable phase: every 3 months
Past psychiatric history		First episode: baseline Recurrent acute episode: baseline with focus on past antipsychotic treatment including type of medication, dose, side effects, response, duration of treatment, medication adherence Stabilization/Stable phases: N/A
Family psychiatric history		First episode: baseline Recurrent acute episode: baseline Stabilization/Stable phases: N/A
Developmental history	Gestational parent's obstetrical history including pre- and perinatal complications, developmental milestones, history of learning disabilities	First episode: baseline Recurrent acute episode: N/A Stabilization/Stable phases: N/A
Medical history		First episode: baseline Recurrent acute episode: baseline Stable phase: yearly
Extrapyramidal side effects	See Table 5 for EPS rating scales	First episode: baseline; when dosage of antipsychotic is changed or new antipsychotic is started, then weekly for 2–4 wk

Parameter	Tools for Screening	Phase of Illness, Recommended Monitoring/Frequency
(akathisia, dystonia, parkinsonism, tardive akathisia, tardive dyskinesia, tardive dystonia)		Recurrent acute episode: baseline; when dosage of antipsychotic is changed or new antipsychotic is started, then weekly for 2–4 wk Stabilization phase: as clinically indicated Stable phase: every clinical encounter or 6 months, or more often for individuals at higher risk
Cognitive functions	Estimates of premorbid/current IQ, attention and concentration, working memory, verbal and visual learning, and memory; executive functions, such as abstract thinking, reasoning, problem solving, judgment	First episode: referral to a psychologist is recommended for neurocognitive testing within 3 months after psychotic symptoms have remitted Recurrent acute episode: N/A Stabilization phase: N/A Stable phase: referral to a psychologist for neurocognitive testing as clinically indicated (clinical evidence of ongoing cognitive impairment that affects functioning)
Functional enquiry and physical examination (with focus on current complaints, endocrine function and sexual function)	Vital signs, weight, BMI, waist circumference	First episode: baseline and then as clinically indicated; weight and BMI monthly for 6 months; baseline waist circumference; blood pressure at baseline and 12 wk or more often if clinically indicated Recurrent acute episode: baseline and then as clinically indicated; weight and BMI monthly for 6 months after initiation of a new antipsychotic; baseline waist circumference; blood pressure at baseline and 12 wk or more often if clinically indicated Stabilization phase: as clinically indicated Stable phase: as clinically indicated; weight and BMI every 3 months when on stable antipsychotic dosage; waist circumference annually; blood pressure annually or more often if clinically indicated; functional enquiry (including endocrine and sexual function) and physical exam at least yearly
Laboratory investigations	CBC/differential, electrolytes, kidney/liver function, fasting glucose and lipid profile, TSH, baseline prolactin, routine urinalysis. If clinically indicated: urine drug screen, tests for STIs, HIV, hepatitis	First episode: baseline; fasting glucose at baseline and repeat at 12 wk or more often as clinically indicated; fasting lipid profile at baseline and 12 wk and repeat as clinically indicated Recurrent acute episode: baseline; fasting glucose at baseline and repeat at 12 wk or more often as clinically indicated; fasting lipid profile at baseline and 12 wk and repeat as clinically indicated Stabilization phase: as clinically indicated Stable phase: fasting glucose annually or more frequently if gaining weight or symptomatic; fasting lipid profile annually or every 6 months if LDL or triglyceride levels above normal range; other tests as clinically indicated

Parameter	Tools for Screening	Phase of Illness, Recommended Monitoring/Frequency
CT brain		First episode: recommended if signs and symptoms are suggestive of intracranial pathology or if onset of symptoms is later in life Recurrent acute episode: as clinically indicated
Genetic testing	Polygenic risk score (i.e., a genetic risk assessment), if available	First episode: consider in first-episode psychosis based on history and physical examination ^[13]

Abbreviations: AUDIT = Alcohol Use Disorders Identification Test; BMI = body mass index; BPRS = Brief Psychiatric Rating Scale; CAGE = Cut-Annoyed-Guilty-Eye substance abuse screening tool; CAGE-AID = CAGE screening tool Adapted to Include Drug use; CBC = complete blood count; CGI-C = Clinical Global Impression Scales for Change; CGI-S = Clinical Global Impression Scales for Severity; CT = computed tomography; EPS = extrapyramidal side effects; HIV = human immunodeficiency virus; IQ = intelligence quotient; LDL = low-density lipoprotein; N/A = not applicable; PANSS = Positive and Negative Syndrome Scale; SOFAS = Social and Occupational Functioning Assessment Scale; STI = sexually transmitted infection; TSH = thyroid-stimulating hormone

Therapeutic Choices

- Patients capable of providing informed consent should be involved in treatment decision-making.^[13]
- Antipsychotic medications are considered the most effective treatment option for patients with schizophrenia and related psychotic disorders,^{[21][22]} but antipsychotic agents need to be integrated with psychosocial interventions to optimize outcomes.^[23]
- Both pharmacologic and psychosocial interventions should be tailored to the individual, based on phase of illness, severity of symptoms, past history of treatment response (if applicable), degree of insight and acceptance of treatment, presence of comorbid psychiatric/physical disorders, and family medical/psychiatric history.

Nonpharmacologic Choices

First Episode/Recurrent Acute Episode

- Determine appropriate treatment setting (least restrictive setting possible), ensure safety, and reduce environmental stressors and stimuli.
- Acutely agitated patients and those at imminent risk of harm to self or others will require hospitalization, if necessary, on an involuntary basis. Criteria for involuntary psychiatric assessment and/or admission are determined by each province's mental-health legislation.
- See patient frequently (for outpatients at least weekly for first 4–6 wk) in order to:
 - build rapport
 - provide support, practical advice and psychoeducation
 - promote medication adherence
 - monitor treatment response (both adverse effects and improvement in symptoms)
- Foster a collaborative, therapeutic relationship between the patient, family/caregivers and treatment team.

Stabilization/Stable Phases^[24]

- In schizophrenia and related psychotic disorders, recovery (stabilization phase) from an acute psychotic episode usually occurs over 6 months but may take longer and may be incomplete. Focus should be on medication adherence, stress management, assessment of signs and symptoms of postpsychotic depression and suicidality, assessment of substance use, and education about early warning signs of relapse.
- Psychosocial interventions are an essential component of treatment in all phases of illness. Individual and family psychoeducation, CBT, motivational interviewing, social and vocational skills training, supported employment programs, and peer support groups have been shown to improve functional outcome and community reintegration, promote treatment adherence, and help prevent relapse.^{[23][25]} Furthermore, CBT may be effective in treating residual psychotic symptoms as well as symptoms of depression and anxiety.^[26]
- Counselling regarding healthy diet and exercise may be helpful in preventing and/or managing antipsychotic side effects.^[20]
- Exercise and tai chi may reduce symptoms and improve motor coordination and memory.^[27]
- Individuals with serious, ongoing illness and functional disability or comorbid problems such as substance abuse may benefit from referral to an Assertive Community Treatment (ACT) team, if available.^[28]
- Maintain continuity of care with an individual clinician or multidisciplinary treatment team.

Pharmacologic Choices

Choice of Antipsychotics

Antipsychotics are effective in the acute and maintenance treatment of schizophrenia and related psychotic disorders.^[29] Two major classes of **antipsychotics** are currently available in Canada: first-generation antipsychotics (FGAs) and second-generation antipsychotics (SGAs).

- All FGAs and SGAs, with the exception of clozapine, have similar efficacy in treating the positive (psychotic) symptoms of schizophrenia and related disorders, and may be considered as initial treatment options;^[30] choice should be guided by tolerability and adverse reaction profile. SGAs are typically considered first due to more favourable tolerability. While some studies have found that SGAs may have advantages in first-episode psychosis^{[31][32]} (improving negative symptoms, mood and cognitive deficits^{[33][34]} and in preventing relapse^[35] and rehospitalization), overall the results have been mixed and greater efficacy for SGAs has not been consistently demonstrated.^{[36][37]} No differences in effect on quality of life have been found between FGAs and SGAs.^[38]
- FGAs, also known as “typical” or “conventional” antipsychotics, can be classified according to their chemical structure (e.g., phenothiazines such as **fluphenazine** or butyrophenones such as **haloperidol**) or potency (low, intermediate, high) as determined by dopamine D₂-receptor binding affinity. When comparing low, intermediate and high-potency agents, differences in side effect profiles are observed (see [Table 3](#) and [Table 6](#)):
 - low-potency agents (e.g., **chlorpromazine**, **methotrimeprazine**) have greater rates of sedation, cardiovascular effects, anticholinergic effects and weight gain
 - intermediate-potency agents (e.g., loxapine, perphenazine and zuclopenthixol)
 - high-potency agents (e.g., **fluphenazine**, **haloperidol**) have greater rates of extrapyramidal side effects (EPS) (e.g., parkinsonism, tardive movement disorders such as tardive dyskinesia), neuroleptic malignant syndrome (NMS) and elevated prolactin levels
- SGAs or “atypical” antipsychotics (**aripiprazole**, **asenapine**, **brexpiprazole**, **cariprazine**, **clozapine**, **lurasidone**, **olanzapine**, **paliperidone**, **quetiapine**, **risperidone**, **ziprasidone**) have greater 5HT_{2A} affinity relative to D₂ affinity. The duration of binding to D₂ receptors and differences in binding affinity to other neurotransmitters (serotonergic, muscarinic, histaminic, alpha-adrenergic) may account for clinical differences in dosing requirements and side effect profiles among SGAs (see [Table 3](#), [Table 4](#) and [Table 7](#)).^[39] Some experts have further classified aripiprazole, brexpiprazole and cariprazine as third-generation antipsychotics due to their unique property of partial dopamine agonism and potential for treating negative symptoms; by having an affinity to dopamine receptors that is lower than endogenous dopamine, they have overall antagonist effects. Asenapine, lurasidone, paliperidone and ziprasidone also have slightly different binding affinities compared with other SGAs; see [Table 7](#) for more information.

Table 3: Basic Pharmacology of Antipsychotics^{[40][41]}

Receptor Antagonized	Clinical Effects/Adverse Effects	Antipsychotics Most Relevant
<i>Note: individual drug actions cannot be predicted by receptor affinities alone, due to complexity of mechanism of action, drug metabolism and brain circuitry. Several adverse effects of APs are not clearly explained by receptor affinities (e.g., metabolic effects, CVD, mortality in dementia, seizures).</i>		
Dopamine (D₂)	Antagonism of D₂ receptors is the basic mechanism of action of all APs, since increased D₂ activity in the mesolimbic tract of the brain may be related to positive symptoms of schizophrenia. Excessive dopamine blockade in other dopaminergic tracts of the brain may be associated with EPS, TD and hyperprolactinemia.	- All APs have D₂ antagonist activity (with the exception of third-generation antipsychotics aripiprazole, brexpiprazole and cariprazine). - FGAs have a higher affinity for D₂ receptors than SGAs and result in more dopamine-related ADRs. - High-potency FGAs have a higher risk of dopamine-related ADRs. - SGAs paliperidone and risperidone can also cause hyperprolactinemia. - Aripiprazole, brexpiprazole and cariprazine are a partial dopamine agonists, but are less potent than endogenous dopamine and result in an overall antagonistic effect.
Histaminic (H₁)	Sedation, appetite stimulation, weight gain.	- More potential: low-potency FGAs, clozapine, quetiapine, olanzapine. - Less potential: aripiprazole, asenapine, brexpiprazole, paliperidone, risperidone, ziprasidone.
Muscarinic (M₁)	Anticholinergic effects (e.g., dry mouth, constipation).	- Low-potency phenothiazines, clozapine, olanzapine, quetiapine. - Avoid in those with significant cognitive impairment, dementia or delirium and in older patients.
Noradrenergic (alpha-1)	Orthostatic hypotension.	- More potential: low-potency FGAs, clozapine, olanzapine, quetiapine, risperidone, ziprasidone. - Less potential: aripiprazole, asenapine, brexpiprazole, lurasidone, paliperidone.
Serotonin (5-HT_{2A})	Antagonism of 5-HT _{2A} receptors may increase dopaminergic activity in the mesocortical pathway (improving negative symptoms of schizophrenia) and the nigrostriatal pathway (decreasing risk of EPS).	- SGAs have significant 5-HT_{2A} antagonist activity. - SGAs have a higher 5-HT_{2A}/D₂ ratio, which maximizes antipsychotic action while minimizing risk of EPS.

Abbreviations: ADR = adverse drug reaction; AP = antipsychotic; EPS = extrapyramidal side effects; FGA = first-generation antipsychotic; SGA = second-generation antipsychotic; TD = tardive dyskinesia

Treatment Considerations

Tailor treatment to the specific phase of the disorder and to the patient's signs and symptoms.^[30]

Acute Phase (First Episode/Recurrent Acute Episode)

The [Figure 1](#) algorithm illustrates the management of patients in the acute phase of a psychotic episode.

- **Haloperidol** IM has been the most widely used treatment for psychotic agitation. The combination of haloperidol and **promethazine** is safe and effective in the management of psychosis-induced aggression.^{[42][43]} Although haloperidol combined with **lorazepam** has been shown to be more effective than haloperidol alone,^[44] this finding was not consistent across multiple studies.^[43]
- While only studied in mildly to moderately agitated patients, **olanzapine** IM appears to have efficacy similar to haloperidol and causes less EPS.^{[45][46]} Do not use the parenteral formulations of benzodiazepines and olanzapine in combination as there have been reports of cardiac and respiratory problems, including death, due to excessive depressive effects.
- Rapid-dissolving oral formulations of olanzapine are as effective as haloperidol IM^[47] if the patient is able and willing to take oral medications.
- **Zuclopenthixol acetate** is an injectable FGA with a pharmacokinetic profile (peak serum level in 24–48 hours, declining to one-third of peak concentration in 72 hours) that may reduce the number of injections required in severe acute agitation and/or aggression. It should not be used in antipsychotic-naïve patients.
- Patients with first-episode psychosis are more responsive to lower doses of antipsychotics,^{[48][49]} have a greater rate of recovery^[48] and are more prone to side effects.
 - When initiating SGAs (with the exception of ziprasidone and extended-release quetiapine), begin with a low dose and titrate gradually over 1–2 weeks up to the usual therapeutic range (see [Table 7](#)).
 - If a FGA is used, consider an intermediate-potency agent, such as **loxapine** or **perphenazine**, to minimize risk of adverse effects.
 - Benzodiazepines can be used to treat anxiety and agitation while titrating the dose of antipsychotic.
- With the exception of ziprasidone and extended-release quetiapine (both are quickly titrated to avoid adverse reactions), rapid titration and high doses of antipsychotics do not accelerate or enhance response and are rarely indicated. Even in patients with a chronic course, 4 mg of haloperidol is as effective as 10–40 mg and associated with fewer side effects and greater tolerability;^[50] in the author's experience, a range of 2–5 mg of haloperidol is usually sufficient. Only a very small subgroup of patients appears to benefit from high-dose therapy.^{[51][52]} Doses exceeding the recommended daily maximum are sometimes required but should be used under the care of a psychiatrist.
- Long-acting injectable antipsychotics (LAIs), such as **aripiprazole**, **paliperidone palmitate** or risperidone, should be offered as an option in all phases of illness, including the first episode, after first establishing tolerability with oral formulations.^[30] See [Long-Acting Injectable Antipsychotics](#) for more information.
- In the first episode, antipsychotic medication should be continued for at least 2 weeks unless there are significant tolerability issues. Assessment of treatment adherence and substance use is necessary in order to determine if there is an actual lack of response. An adequate trial of antipsychotic therapy is 4–6 weeks at a dose within the usual therapeutic range.^{[30][34][53]} If there is a partial response at 4 weeks, reassess at 8 weeks in order to determine if a switch to a different antipsychotic medication is indicated. Patients who fail to demonstrate even minimal response by that time are unlikely to benefit from a longer trial; consider switching to a different antipsychotic and consulting a psychiatrist.

Stabilization/Stable Phases

- During the stabilization (recovery) phase, patients are vulnerable to relapse. Avoid changes in antipsychotic medication unless there are intolerable side effects or persistent residual symptoms that are distressing and/or disabling to the patient.
- Maintenance therapy is essential to prevent relapse. First-episode psychosis is associated with a 70–90% risk of relapse within 5 years. A significant percentage of patients relapse as early as the second year following discontinuation of antipsychotic therapy for first-episode psychosis.^[54]
 - Continue maintenance pharmacotherapy for at least 1–2 years for first-episode patients who achieve symptom remission and functional recovery.^{[30][53][55]} Longer treatment (2–5 years) may be required for individuals with a long duration of untreated psychosis, more severe illness, slower response, substance abuse, and history of suicidal or aggressive behaviour.^{[30][53]} For patients with a history of 2 or more episodes, continue maintenance pharmacotherapy until the patient has been stable and relapse-free for at least 5 years.^[30]
 - Many patients will require antipsychotic treatment indefinitely.
 - In general, the lowest effective antipsychotic dose used during the acute phase should also be used in maintenance treatment.^[21]^[30] Reducing the antipsychotic dose in the treatment of multi-episode schizophrenia is not recommended, as this is associated with increased rates of relapse.^[56] There is little evidence to support the use of more than 1 antipsychotic in acute or maintenance treatment for the majority of patients.^[57]
- Poor medication adherence is common in patients with schizophrenia and related psychotic disorders. Adherence rates of <70–80% are associated with a significantly increased risk of relapse and hospitalization.^{[58][59]} Patients who discontinue medications are 5 times more likely to relapse.^[22] Recovery from recurrent acute psychotic episodes (relapses) may take longer with each subsequent episode, and the degree of recovery may not be as great, resulting in persistent residual symptoms and functional disability.^[22]
- When discontinuing medication, gradually reduce the dose by ≤20% every 2–4 weeks. Reduce over a period of 6–12 months for first-episode patients and 6–24 months if patients have experienced 2 or more episodes.^[21] Monitor closely; if patients experience early signs of relapse, restabilize them on the previously effective dose of antipsychotic as quickly as possible.

Long-Acting Injectable Antipsychotics

- Long-acting injectable antipsychotics, such as **aripiprazole**, **paliperidone palmitate** or **risperidone**, should be offered as an option in all phases of illness, including the first episode, after first establishing tolerability with oral formulations.^[30]
- In addition to promoting adherence, LAIs may improve rates of remission and decrease the risk of hospitalization and relapse.^[60]^[61] A recent RCT in first-episode schizophrenia found better symptom control and a 6-fold reduction in relapse at 1 year with LAIs compared to oral formulations.^[30]
- LAI usage rates in Canada are low (6.3%) compared with the rest of the world (15–80%).^[61] Canadian qualitative studies of physician and patient attitudes toward LAIs indicate that physicians' lack of knowledge and training in the use of LAIs, negative biases toward LAIs (treatment of last resort) and unfounded beliefs regarding patients' rejection of LAIs due to fear of needles may contribute to the low rates of use.^{[62][63]}

Antipsychotic Polypharmacy

- Combination therapy with more than 1 antipsychotic medication (antipsychotic polypharmacy) is not supported by available scientific evidence except for:^[20]
 - cross-titration when switching to a different antipsychotic
 - adjunctive treatment for partial responders to clozapine
- There is no good evidence to guide switching from one antipsychotic medication to another. Crossover medication strategies, over 2 weeks to 3 months,^[64] are preferred. **SwitchRx** is a Canadian online medication switching tool that provides clinicians with guidance on how to transition patients from one antipsychotic to another. It is important to complete the crossover to avoid unnecessary polypharmacy.
- Antipsychotic polypharmacy can lead to increased risk of drug-drug interactions that can be life-threatening, decreased efficacy, increased side effects and decreased adherence to treatment. It is therefore recommended that antipsychotic polypharmacy be

used only in exceptional circumstances under the care of a psychiatrist.^[65]

Treatment-Resistant Schizophrenia

Treatment-resistant schizophrenia is commonly defined as a failure to respond (≤20% reduction in positive symptoms) to 2 or more adequate trials (≥6 wk) of non-clozapine antipsychotic medication.^{[30][66]} **Clozapine** is the only antipsychotic with proven efficacy in treatment-resistant schizophrenia^[21] in reducing hostility and aggression,^{[67][68]} persistent suicidality, and all-cause mortality.^[69] Patients successfully stabilized on clozapine have not been shown to benefit further from switching to any other antipsychotic medication or from the addition of a second antipsychotic.^[30]

Clozapine is reserved for treatment-resistant schizophrenia due to the risk of agranulocytosis and the need for regular blood monitoring. A discussion on the assessment and management of treatment-resistant schizophrenia is beyond the scope of this chapter, but may be found in the Canadian schizophrenia guidelines (see [Suggested Readings](#)).

Treatment of Comorbid Conditions

Depression and Suicidality

- Psychiatric comorbidities are common among individuals with schizophrenia with an estimated prevalence of 50% for depression, 29% for post-traumatic stress disorder, 23% for obsessive-compulsive disorder and 15% for panic disorder.^[70] The lifetime risk of suicide is approximately 5%.^[71]
- Depressive symptoms are common in the prodromal phase preceding the onset of a first episode of psychosis. In the acute phase, particularly in multiple-episode patients, depressive symptoms usually remit along with the positive psychotic symptoms.
 - SGAs may be more effective than FGAs in treating depressive symptoms.^[72] In the acute phase. In those with persistent suicidality, consideration should be given to clozapine.^[24] There is minimal evidence to support the use of an antidepressant in the acute phase.^[30]
- Major depressive episodes occur as often in individuals diagnosed with schizophrenia as in those with either schizoaffective disorder or nonpsychotic major depression.^[73]
 - Patients with first-episode psychosis have a greater risk of depression compared with multiple-episode patients, particularly in the stabilization phase (postpsychotic depression);^[73] depressive symptoms tend to increase for the first 3 months following a first episode of psychosis.
- It can be difficult to differentiate between depression and ongoing negative symptoms or antipsychotic-induced emotional blunting. Rating scales, such as the Hamilton Rating Scale for Depression (HAM-D) and the Beck Depression Inventory, have not been validated in individuals with schizophrenia. The Calgary Depression Scale for Schizophrenia (CDSS)^[74] can be used to differentiate between negative symptoms and symptoms of major depression. A CDSS score ≥6 is considered clinically significant.^[75]
- Antidepressant medication may be useful in the treatment of major depression in the stabilization or stable phases.^[76] As stated in [Nonpharmacologic Choices](#), CBT may be effective in treating residual psychotic symptoms as well as symptoms of depression and anxiety.^[26]

Substance Abuse

- Higher rates of substance abuse/dependence disorders are found in individuals with schizophrenia (lifetime prevalence of 47–50%) compared with the general population. Persistent substance abuse is associated with poor treatment adherence, greater risk of suicidality and aggression, and significantly poorer outcomes.
- **Cannabis** use is a risk factor for psychosis and there appears to be a dose-response relationship between amount of use and risk of psychosis;^{[77][78]} higher-potency cannabis products may also confer increased risk.^[79] The prevalence of cannabis use in individuals with psychotic disorders is higher than in the general population, particularly in early psychosis where rates of use are as high as 86% and rates of cannabis abuse/dependence range from 14–28%.^[80] Cannabis use is associated with poor antipsychotic adherence, greater severity and chronicity of symptoms, and 4 times the risk of relapse.^[81]

- The rate of smoking in individuals with schizophrenia and related psychotic disorders is extremely high (70% in Canada).^[82] compared with a rate of approximately 20% in the general population (www.smoke-free.ca). Patients with psychotic disorders are more likely to smoke heavily and are 2–3 times more likely than those who don't smoke to abuse other substances.^[83]
 - It is estimated that patients with schizophrenia have a life expectancy that is 20% lower than the general population. This is mostly attributable to increased rates of cardiovascular disease and metabolic syndrome, for which smoking is a significant risk factor.^[84]
 - Polycyclic aromatic hydrocarbons found in cigarette smoke can induce the metabolism of some antipsychotics (e.g., clozapine and olanzapine); people who smoke require higher doses of these antipsychotics, leading to increased adverse effects.^[85] Dose adjustments may be required during hospitalization, when patients stop and restart smoking. These interactions are not related to nicotine; no further adjustments are required if nicotine replacement therapy is used.
 - Actively encourage smoking cessation, even though success rates are low in patients with psychotic disorders (in patients with psychoses, only 9% are people who formerly smoked compared with 14–49% in the general population).^[82] Nicotine replacement therapies may be helpful. **Bupropion** and **varenicline** are also effective in smoking cessation; the risks of new or worsening neuropsychiatric symptoms have been shown to be lower than previously reported.^[86] However, monitoring for changes in mood and behaviour should still occur, especially in those with a history of suicidal ideation/attempts, depressed mood or agitation/hostility. For more information, see [Tobacco Use Disorder: Smoking and Nicotine Cessation](#).
- In patients (and particularly in those with first-episode psychosis) who appear to have been using substances as a form of self-medication, substance abuse may remit spontaneously with recovery from the acute episode.
- A harm-reduction approach that incorporates motivational interviewing is recommended for individuals with comorbid substance abuse problems. Refer patients with persistent substance abuse to a mental-health program that can provide a comprehensive, integrated approach.^[87]

Side Effects of Antipsychotics

[Table 4](#) compares the side effect profile of second-generation antipsychotics. [Table 5](#) provides guidance on the assessment and management of antipsychotic-induced side effects.

- Closely monitor and manage side effects, since they are the leading cause of nonadherence to medication.
- Patients taking antipsychotics long term ranked the most burdensome and function-impairing side effects as follows: sedation/tiredness, weight gain, difficulty thinking/concentrating (often due to sedation) and restlessness (an EPS).^[88]

Table 4: Comparative Side Effects of Second-Generation Antipsychotics

Medication ^[40]	Sedation	Insomnia	EPS ^[a]	Weight Gain	Metabolic Abnormalities ^[b]	Hyperprolactinemia	CV Effects ^[c]
Aripiprazole	+/-	++	++	+	+/-	+/-	+
Asenapine	+/-	+/-	+	+/-	+/-	+/-	+
Brexpiprazole	+/-	+/-	-	+	+/-	+/-	+/-
Clozapine	++++	-	-	++++	++++	-	++
Lurasidone	+	+/-	+	++	+/-	+/-	+
Olanzapine	+++/>++	-	+/-	++++	++++	+	+

Medication ^[40]	Sedation	Insomnia	EPS ^[a]	Weight Gain	Metabolic Abnormalities ^[b]	Hyperprolactinemia	CV Effects ^[c]
Paliperidone	+/-	++	+++/>++	++/>+	+/-	+++/>++	+
Quetiapine	+++	-	-	+++	++	-	+
Risperidone	+	+	+++	++	+/-	+++	+
Ziprasidone	+	+	+	+/-	+/-	+	++

^[a] Includes akathisia.
^[b] Includes glucose abnormalities and dyslipidemias.
^[c] Includes orthostatic hypotension and/or QTc prolongation.
Legend: ++++ = high; +++ = moderate; ++ = low; + = very low; +/- = minimal or none; - = equivalent to placebo
Abbreviations: CV = cardiovascular; EPS = extrapyramidal side effects

Table 5: Antipsychotic-Induced Side Effects: Assessment, Monitoring and Management^{[20][89][90]}

Side Effects	Incidence, Assessment and Monitoring	Management
Cardiovascular	Incidence: orthostatic hypotension is the most common AP-induced CVD effect, usually with low-potency FGAs and clozapine, but can occur with other SGAs. QTc prolongation may occur with chlorpromazine, pimozide and haloperidol. Among SGAs, ziprasidone has a greater (though modest) capacity for QTc prolongation. APs confer a dose-related increase in the risk of sudden cardiac death^[91] and an increased risk of MI.^[92] Clozapine is associated with an increased risk of myocarditis (especially in, but not limited to, the first month of therapy) and other CVD complications. Assessment: baseline vital signs; family history and medical history. In individuals at risk of electrolyte disturbance, particularly hypokalemia and/or hypomagnesemia (e.g., history of kidney disease, diuretic therapy, water intoxication, eating disorders, prolonged vomiting/diarrhea, alcoholism), measure baseline electrolytes and correct abnormalities. The 2021 APA guidelines recommend baseline ECG in patients with cardiovascular risk factors for QTc prolongation and those starting chlorpromazine, pimozide or ziprasidone.^[24] Monitoring: vital signs at 12 wk then annually; monitor more frequently as indicated and when initiating clozapine until dose is stable; annual cardiovascular risk assessment. ECG as clinically indicated, with significant dose changes (chlorpromazine, pimozide or ziprasidone),	• Educate about risks and prevention of orthostatic hypotension. • If symptoms persist, decrease dose of AP, if possible, or switch to another AP. • Clozapine-induced myocarditis: discontinue clozapine immediately and consult cardiology. • Despite these and other contributory CVD risk factors (e.g., metabolic), APs reduce overall mortality risk.^[93]

Side Effects	Incidence, Assessment and Monitoring	Management
	with addition of other medications that can affect QT _c in patients with cardiovascular risk factors for QT _c prolongation or elevated baseline levels. ^[24] For clozapine: troponin, CRP and ECG in presence of signs and symptoms of possible myocarditis.	
Dermatologic	Incidence: DRESS (rare) associated with several SGAs.^[94] Mortality rates of up to 10% have been reported.^[95] Assessment: rash, fever, swollen lymph nodes and swollen face; can result in injury to liver, kidneys, lungs, heart and pancreas.	Early recognition, stop AP, supportive care (see Drug-Induced Skin Reactions).
Dyslipidemias	Incidence: hyperlipidemia has been associated with clozapine and olanzapine. Increased triglycerides have been reported with quetiapine. Assessment: baseline fasting lipid profile (total cholesterol, LDL, HDL, triglycerides). Monitoring: fasting lipid profile at 12 wk after initiation of a new AP, then annually; more frequent monitoring if significant weight gain, and every 6 months if LDL and/or triglycerides above the normal range.	- Educate on change in diet. Consult the Canadian Cardiovascular Society guidelines for the management of dyslipidemia. - Consider switching to ziprasidone^{[96][97]} or another agent with lower risk (see Table 4).
Endocrine, sexual and urinary effects	Incidence: hyperprolactinemia is a common side effect of high-potency FGAs, risperidone and paliperidone. Sexual dysfunction is associated with all APs and is not necessarily related to elevated prolactin levels alone. Urinary retention risk is greatest with drugs that have significant anticholinergic activity.^[98] Assessment: baseline functional inquiry including libido, menstrual history, erectile and ejaculatory function; baseline prolactin level in first-episode psychosis, before initiating an AP associated with hyperprolactinemia and when indicated. Monitoring: monthly for 3 months after initiating a new AP, then yearly.	- Determine underlying cause of endocrine or sexual dysfunction and treat accordingly. - Consider drugs for erectile dysfunction. - For clinically significant hyperprolactinemia, reduce dose of AP; if dose reduction not tolerated (emergence of or increase in psychotic symptoms), consider switching to an AP not associated with hyperprolactinemia (see Table 4).
Extrapyramidal side effects (akathisia, dystonia, parkinsonism, tardive akathisia, tardive	Incidence: SGAs have a significantly reduced risk of EPS and tardive dyskinesia compared with FGAs.^{[21][33][34][99][100]} The incidence of TD is 5.5% per year with FGAs and 3.9% with SGAs.^[20] Assessment: ESRS is the recommended tool to assess EPS including tardive dyskinesia. Other tools include AIMS, BARS and SAS. Parkinsonism can be associated with dysphoria, decreased concentration and slowing of	- Prevention is key—use SGAs first line. If EPS occurs, reduce dose; consider switch to SGA if on FGA.^[24] If EPS persists, consider anticholinergics (benztropine, procyclidine, trihexyphenidyl). Prophylactic use of anticholinergics is not recommended. - Akathisia: if dose reduction is not effective, beta-blockers (e.g., propranolol

Side Effects	Incidence, Assessment and Monitoring	Management
dyskinesia, tardive dystonia)	cognition. Akathisia (subjective and objective restlessness) is often misidentified as psychotic agitation, which can result in an increase in dose of the offending AP. Tardive dyskinesia consists of repetitive, involuntary choreoathetoid movements usually involving the buccal-oral-lingual musculature, face, trunk, extremities or respiratory muscles, and can be permanent and disabling. Monitoring: baseline assessment in AP-naïve first-episode patients, in multiple-episode patients when initiating a new AP, and in first-episode and multiple-episode patients whenever dosage of AP is changed. Monitor weekly for 2–4 wk or until EPS resolves; in stable patients, monitor for TD at every clinical encounter or every 6 months or more often in patients at higher risk (on FGAs, erratic medication adherence or intermittent treatment, female,^[a] age >55, diagnosis of an affective disorder, substance abuse, diabetes).	10–120 mg/day) are treatment of choice with monitoring for hypotension. Benzodiazepines also provide symptom relief. Anticholinergics are ineffective. - Acute dystonia (acute torticollis, oculogyric crisis): benztropine or diphenhydramine IM followed by reduction in dose or switch to SGA. - Tardive dyskinesia: no evidence-based treatment; therefore, prevention is key.^[101] Use SGAs first line. Antiparkinsonian medications not effective and may worsen symptoms. If TD occurs, suggest consultation with a psychiatrist. Consider switching to an SGA. For persistent, severe TD, consider clozapine trial. There is limited evidence for use of tetrabenazine but it may be considered if TD persists.^[24]
Glucose abnormalities	Incidence: both FGAs and SGAs are associated with greater risk of glucose abnormalities including hyperglycemia, insulin resistance, new onset type 2 diabetes, exacerbation of type 1 diabetes and diabetic ketoacidosis.^[102] Among SGAs, clozapine and olanzapine have the highest risk. Assessment: baseline fasting plasma glucose; HbA_{1c} if difficult to obtain fasting plasma glucose; OGTT if evidence of impaired glucose tolerance; family history and medical history. Monitoring: signs and symptoms of emergent diabetes; fasting plasma glucose 12 wk after initiating new AP, then yearly; more frequent monitoring required if significant weight gain or if symptomatic; OGTT if evidence of impaired glucose tolerance.	- Educate about signs and symptoms of emergent diabetes. If diabetes is diagnosed, follow the Diabetes Canada guidelines for diabetes and mental health. - Consider switching to another AP with lower risk (see Table 4).
Neuroleptic malignant syndrome	Incidence: rare, but can occur with any AP at any dose and at any time. Assessment: physical exam with focus on level of consciousness; vital signs (fever, tachycardia, fluctuations in BP); look for evidence of severe muscle rigidity (not to be confused with cogwheel-type rigidity due to drug-induced parkinsonism) and elevated levels of creatine kinase and white blood cells. Risk factors include young age, male,^[a] neurologic disabilities, dehydration, agitation, exhaustion, and rapid or parenteral administration of AP.	- Medical emergency: discontinue AP and provide supportive care (hydration and cooling). - Dantrolene 2–3 mg/kg TID–QID IV (maximum 10 mg/kg daily) or bromocriptine 2.5–5 mg TID PO, increasing by 2.5 mg TID Q24H (maximum 60 mg daily).^[64]

Side Effects	Incidence, Assessment and Monitoring	Management
Sedation and cognitive effects	Incidence: sedation is very common and can be disabling, especially when first starting AP or increasing dose. Low-potency FGAs and some SGAs are implicated. Cognitive effects are less common with SGAs, but can occur especially with SGAs with greater anticholinergic activity (e.g., clozapine, olanzapine, quetiapine). Assessment: daytime drowsiness, excessive sleep, cognitive “dulling”; obtain collateral information from caregivers and family; anticholinergics (for EPS) and antiepileptics (for mood stabilization) can worsen cognitive effects.^{[105][106]} Monitoring: weekly for first 4–6 wk after initiating new AP.	• Use SGAs as first-line treatment to reduce risk of cognitive side effects.^{[103][104]} • Use a low initial dose and gradual titration based on degree of sedation, especially with clozapine and first-episode psychosis (exceptions include quetiapine XR and ziprasidone, for which rapid titration is recommended). • Give entire daily dose at HS if possible to reduce daytime drowsiness. • Do not use anticholinergic antiparkinsonian agents prophylactically; avoid prolonged use when treating acute EPS.
Weight gain	Incidence: for FGAs, the risk of weight gain is greatest with low-potency agents. For many SGAs, there is a higher risk of weight gain compared with FGAs (see Table 3); may be rapid and usually plateaus within 1 y. Assessment: baseline weight, body mass index (BMI), waist circumference. Monitoring: weight and BMI monthly for 6 months with new AP then every 3 months when on a stable dosage; waist circumference annually.	• Prevention is critical, as weight reduction is especially difficult in individuals with mental illness. Educate about risk of weight gain and provide dietary and exercise counselling. • If weight increases >7% over baseline, implement behavioural weight reduction program. • If unsuccessful, assess risks/benefits of continuing current AP vs. switching to an AP with lower risk (see Table 4). • In CATIE, ziprasidone was the only agent to show improvement in weight and metabolic parameters following a switch from another SGA.^[96] • Dose reduction unlikely to result in weight loss. • Addition of metformin can also be considered.^[20]

Abbreviations: AIMS = Abnormal Involuntary Movement Scale; AP = antipsychotic; BARS = Barnes Akathisia Rating Scale; BMI = body mass index; BP = blood pressure; CATIE = Clinical Antipsychotic Trials of Intervention Effectiveness; CRP = C-reactive protein; CVD = cardiovascular disease; DRESS = drug reaction with eosinophilia and systemic symptoms; ECG = electrocardiogram; EPS = extrapyramidal side effects; ESRS = Extrapyramidal Symptom Rating Scale; FGA = first-generation antipsychotic; HDL = high-density lipoprotein; LDL = low-density lipoprotein; MI = myocardial infarction; OGTT = oral glucose tolerance test; SAS = Simpson-Angus Scale; SGA = second-generation antipsychotic; TD = tardive dyskinesia

^[a] Studies presenting this finding classify patients as male or female. CPhA is aware these terms may not reflect the identity of the individual patient presenting and are not inclusive of all patients. Each patient should be considered individually.

Choices during Pregnancy and Breastfeeding

Prenatal and Perinatal Risk Factors for Schizophrenia

A number of prenatal and perinatal factors have been associated with an increased risk of schizophrenia in offspring to varying degrees. These include malnutrition, viral illnesses such as influenza, obesity, x-ray radiation, exposure to cat feces infected with *Toxoplasmosis gondii* protozoa, use of analgesics (e.g., ASA), season of birth (lack of sunlight exposure/vitamin D deficiency), maternal-fetal Rh incompatibility, low birth weight and obstetric complications (hypoxia).

Psychosis and Pregnancy

The onset of a first episode of psychosis during pregnancy is uncommon but if it occurs, it is considered a psychiatric emergency and requires treatment because of the potential adverse effects of psychosis on both the pregnant patient and the fetus. Patients who experience a psychotic episode during pregnancy have twice the rate of adverse outcomes including stillbirth, infant death, prematurity and small size for gestational age, even when controlling for other variables including age, education, smoking, marital status, parity and pregnancy-induced hypertension.^[107]

Although patients with schizophrenia may have lower fertility rates compared with the general population, their relative fertility has increased since deinstitutionalization, possibly due to a change in drug selection. High-potency antipsychotics are prescribed less often or at lower doses, resulting in less effect on prolactin levels and ovulation inhibition. More recent studies have found similar numbers of pregnancies in patients with schizophrenia compared with the general population.^{[108][109]} Patients with schizophrenia also have higher rates of unplanned and unwanted pregnancies; less social support; poorer antenatal care; poorer nutrition; greater use of tobacco, alcohol and illicit substances; and the majority experience intermittent loss of custody.^[110]

Patients with schizophrenia and related psychotic disorders should receive counselling about contraception and pregnancy early in the course of their illness. If a pregnancy is planned, a consultation with a psychiatrist is recommended to evaluate the risks/benefits of maintenance antipsychotics during pregnancy. Pregnancy in a patient being treated for schizophrenia is considered high-risk; referrals to a mental-health clinic and obstetrician are recommended and, if possible, the pregnancy should be managed through a high-risk clinic.

Psychosis and the Postpartum Period

The rate of postpartum psychosis is 1–2/1000 births; it usually occurs within 2–4 weeks after delivery. This is a psychiatric emergency because of the rare but serious risk of suicide and/or infanticide. If hospitalization is required and the patient's condition does not pose a risk to the newborn, the patient and baby should be kept together, ideally in a specialized unit.

In the postpartum period, the risk of relapse of schizophrenia and related psychosis is high (~24%) and is greatest during the first 3 months. Patients with schizoaffective disorder are particularly at risk.^[111] If an individual has a history of postpartum psychosis, the overall risk of relapse in a subsequent pregnancy is approximately 35%.^[112]

Antipsychotics and Pregnancy

There have been no randomized controlled trials of FGAs or SGAs in pregnant patients. Teratogenic effects, including major congenital malformations, have not been demonstrated with the use of antipsychotics during pregnancy.^[113] A large cohort study demonstrated increased risk of withdrawal symptoms, neurological disorders and persistent pulmonary hypertension among infants exposed to antipsychotics during pregnancy; absolute incidence remained low (<2%).^[114] In terms of long-term effects, another large cohort study did not find increased risk of neurodevelopmental disorders (ADHD, autism) with antipsychotic use during pregnancy.^[115] A meta-analysis found that patients who continued antipsychotics during pregnancy had an increased risk of gestational diabetes; olanzapine and quetiapine may be particularly problematic due to their metabolic adverse effects.^[116]

While minimizing exposure to all medications is generally advisable in pregnancy, especially during the first trimester, the risks/benefits of antipsychotic use and the choice of medication must be carefully weighed for each patient, considering factors such as severity of past psychotic episodes, vulnerability to relapse, remission versus ongoing residual symptoms, current substance use, level of functioning, availability of social supports and history of prior response including side effects (EPS, weight gain, diabetes).

For many pregnant patients with schizophrenia and related psychotic disorders, maintenance antipsychotics at the lowest dose possible to prevent relapse will be recommended, as maintenance therapy may ultimately reduce risks to the fetus associated with an acute psychotic relapse, such as exposure to higher maternal doses of medication.^{[110][117]}

Antipsychotics and the Postpartum Period

Because of the high risk of relapse in the postpartum period, restart antipsychotics immediately after the delivery if they were discontinued during pregnancy. Closely monitor patients maintained on low-dose maintenance antipsychotics during pregnancy for early signs of relapse, as an increased dose may be needed in the postpartum period. In infants exposed to clozapine in utero, a neutrophil count is recommended; in those exposed to high-potency antipsychotics in utero (third trimester), monitor for EPS for several days.^[73]

Antipsychotics and Breastfeeding

Antipsychotics are generally passed into breast milk and are detectable at levels considered compatible with breastfeeding; however, data in this patient population is limited, and long-term studies are needed.^[118] With use of FGAs, drowsiness and lethargy have been reported, although many studies cite no adverse events.^[118] Of the SGAs, olanzapine is the preferred agent in breastfeeding patients based on over 100 documented cases of low infant exposure.^[119] Quetiapine also demonstrated low infant exposure, although the number of documented cases was lower than olanzapine. Risperidone was associated with more infant exposure than other SGAs and may not be a preferred option. Breastfeeding while taking clozapine is not recommended.^[73]

There is no consensus regarding breastfeeding in patients being treated with antipsychotics; some experts recommend against breastfeeding until further data are available.^[110] The potential benefits of breastfeeding must be weighed against potential adverse effects and discussed with each individual patient. If antipsychotics are used, monitor the infant for adverse effects such as poor feeding, lethargy, drowsiness and delayed developmental milestones.

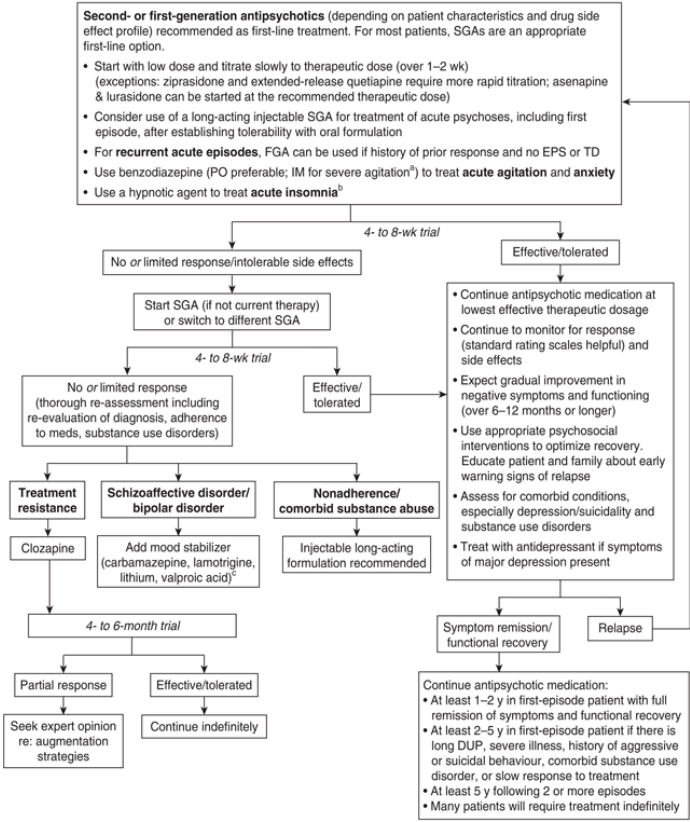
A discussion of general principles on the use of medications in these special populations can be found in [Drug Use During Pregnancy](#) and [Drug Use During Breastfeeding](#). Other specialized reference sources are also provided in these appendices.

Therapeutic Tips

- Early detection of possible psychosis, including recognition of prodromal psychosis and referral to a specialized early psychosis program, is associated with better symptomatic and functional outcomes.
- Initiate treatment before the development of a crisis, such as self-harm, aggression or violence.^[53]
- Provide treatment in the least restrictive setting, considering safety issues, availability of community resources (including both caregiver and mental-health supports), the patient’s insight and competency to consent to treatment, and ability to cooperate with treatment.^[53]
- Integrate psychosocial interventions, such as patient and family psychoeducation, supportive therapy, motivational interviewing, and stress management, with the use of antipsychotic medications in order to promote adherence to treatment and optimize outcomes.^{[23][53]}
- Conduct baseline and regular ongoing assessments of signs and symptoms, possible comorbid conditions (e.g., depression, anxiety, substance use), risk of suicide and aggression, level of functioning, response to treatment, medication side effects, and adherence during all phases of the disorder. A variety of standardized scales and semistructured interviews facilitate these assessments (see [Table 2](#)).^[13]
- To ensure continuity of care, longitudinal follow-up by the same clinician or multidisciplinary team is optimal.^[53]

Algorithms

Figure 1: Management of Acute Psychotic Episodes



[a] Should not be administered simultaneously with parenteral olanzapine due to reports of cardiac and respiratory problems including deaths.

[b] See [Insomnia](#).

[c] Valproic acid should be avoided in patients of childbearing potential. Lithium and carbamazepine are also associated with congenital malformations.

Abbreviations: DUP = duration of untreated psychosis; EPS = extrapyramidal side effects; FGA = first-generation antipsychotic; SGA = second-generation antipsychotic; TD = tardive dyskinesia

Drug Tables

Table 6: Drug Therapy for Psychosis: First-Generation Antipsychotics

Drug/Cost ^[a]	Dosage ^[b]	Adverse Effects ^[c]	Drug Interactions	Comments
Drug Class: Antipsychotics, first-generation, low potency				
chlorpromazine generics	Initial: 50–100 mg/day PO	More common with low-potency agents: sedation, cardiovascular effects, anticholinergic effects, weight gain, lower seizure threshold, photosensitivity. More common with high-potency agents: increased prolactin, EPS , NMS , tardive movement disorders. Liver function abnormalities.	Additive effects with other CNS depressants, anticholinergics, alpha-adrenergic antagonists; inhibitors of cytochrome P450 enzymes (e.g., TCAs , fluoxetine, fluvoxamine, paroxetine) may increase serum levels; inducers of cytochrome P450 enzymes (e.g., carbamazepine, phenytoin) may decrease serum levels; effects of levodopa may be inhibited.	Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.
\$25–50	Usual: 200–400 mg/day PO			
	Maximum: 1000–2000 mg/day; limited benefits at doses >1000 mg/day			
	Divided in 1–4 doses/day			
methotrimeprazine generics	Initial: 25–75 mg/day PO	More common with low-potency agents: sedation, cardiovascular effects, anticholinergic effects, weight gain, lower seizure threshold, photosensitivity. More common with high-potency agents: increased prolactin, EPS , NMS , tardive movement disorders. Liver function abnormalities.	Additive effects with other CNS depressants, anticholinergics, alpha-adrenergic antagonists; inhibitors of cytochrome P450 enzymes (e.g., TCAs , fluoxetine, fluvoxamine, paroxetine) may increase serum levels; inducers of cytochrome P450 enzymes (e.g., carbamazepine, phenytoin) may decrease serum levels; effects of levodopa may be inhibited.	Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.
\$25–50	Usual: 50–200 mg/day PO			
	Maximum: 1000 mg/day			
	Divided in 1–3 doses/day			
methotrimeprazine injection Nozinan	75–100 mg IM given as 3–4 deep injections in a large muscle	More common with low-potency agents: sedation, cardiovascular effects,	Additive effects with other CNS depressants, anticholinergics, alpha-adrenergic antagonists; inhibitors of cytochrome	Advise patients about antipsychotic-associated body temperature dysregulation and

Drug/Cost ^[a]	Dosage ^[b]	Adverse Effects ^[c]	Drug Interactions	Comments
		anticholinergic effects, weight gain, lower seizure threshold, photosensitivity. More common with high-potency agents: increased prolactin, EPS , NMS , tardive movement disorders. Liver function abnormalities.	P450 enzymes (e.g., TCAs , fluoxetine, fluvoxamine, paroxetine) may increase serum levels; inducers of cytochrome P450 enzymes (e.g., carbamazepine, phenytoin) may decrease serum levels; effects of levodopa may be inhibited.	prevention of heat stroke, e.g., hydration, sun protection. IM administration may be used to provide symptomatic control in patients with acute agitation or those who are unable to take the drug orally. Switch to the oral route as soon as practical, due to greater risk of EPS with IM administration.
Drug Class: Antipsychotics, first-generation, intermediate potency				
loxapine Xylac	Initial: 10–20 mg/day PO	More common with low-potency agents: sedation, cardiovascular effects, anticholinergic effects, weight gain, lower seizure threshold, photosensitivity. More common with high-potency agents: increased prolactin, EPS , NMS , tardive movement disorders. Liver function abnormalities.	Additive effects with other CNS depressants, anticholinergics, alpha-adrenergic antagonists; inhibitors of cytochrome P450 enzymes (e.g., TCAs , fluoxetine, fluvoxamine, paroxetine) may increase serum levels; inducers of cytochrome P450 enzymes (e.g., carbamazepine, phenytoin) may decrease serum levels; effects of levodopa may be inhibited.	Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.
\$25–50	Usual: 60–100 mg/day PO; some patients can be maintained on 20–60 mg/day			
	Maximum: 250 mg/day			
	Divided in 2–3 doses/day			
loxapine injection Loxapac	12.5–50 mg IM every 4–6 h or longer; BID dosing often sufficient	More common with low-potency agents: sedation, cardiovascular effects, anticholinergic effects, weight gain, lower seizure threshold, photosensitivity. More common with high-potency agents:	Additive effects with other CNS depressants, anticholinergics, alpha-adrenergic antagonists; inhibitors of cytochrome P450 enzymes (e.g., TCAs , fluoxetine, fluvoxamine, paroxetine) may increase serum levels; inducers of cytochrome P450 enzymes (e.g., carbamazepine, phenytoin) may decrease	Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection. IM administration may be used to provide symptomatic control in

Drug/Cost [a]	Dosage [b]	Adverse Effects [c]	Drug Interactions	Comments
		increased prolactin, EPS , NMS , tardive movement disorders. Liver function abnormalities.	serum levels; effects of levodopa may be inhibited.	patients with acute agitation or those who are unable to take the drug orally. Switch to the oral route as soon as practical, due to greater risk of EPS with IM administration.
perphenazine generics < \$25	Initial: 4–12 mg/day PO Usual: 12–24 mg/day PO Maximum: 48–64 mg/day Divided in 2–3 doses/day	More common with low-potency agents: sedation, cardiovascular effects, anticholinergic effects, weight gain, lower seizure threshold, photosensitivity. More common with high-potency agents: increased prolactin, EPS , NMS , tardive movement disorders. Liver function abnormalities.	Additive effects with other CNS depressants, anticholinergics, alpha-adrenergic antagonists; inhibitors of cytochrome P450 enzymes (e.g., TCAs , fluoxetine, fluvoxamine, paroxetine) may increase serum levels; inducers of cytochrome P450 enzymes (e.g., carbamazepine, phenytoin) may decrease serum levels; effects of levodopa may be inhibited.	Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.
zuclopenthixol Clopixol \$50–75	Initial: 10–50 mg/day divided in 2–3 doses PO Usual: 20–60 mg/day PO Maximum: 100 mg/day Divided in 1–3 doses/day	More common with low-potency agents: sedation, cardiovascular effects, anticholinergic effects, weight gain, lower seizure threshold, photosensitivity. More common with high-potency agents: increased prolactin, EPS , NMS , tardive movement disorders. Liver function abnormalities.	Additive effects with other CNS depressants, anticholinergics, alpha-adrenergic antagonists; inhibitors of cytochrome P450 enzymes (e.g., TCAs , fluoxetine, fluvoxamine, paroxetine) may increase serum levels; inducers of cytochrome P450 enzymes (e.g., carbamazepine, phenytoin) may decrease serum levels; effects of levodopa may be inhibited.	Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.

Drug/Cost [a]	Dosage [b]	Adverse Effects [c]	Drug Interactions	Comments
zuclopenthixol acetate injection Clopixol Acuphase < \$25/50 mg vial	50–150 mg IM Q2–3 days Maximum: cumulative dose of 400 mg or 4 injections	More common with low-potency agents: sedation, cardiovascular effects, anticholinergic effects, weight gain, lower seizure threshold, photosensitivity. More common with high-potency agents: increased prolactin, EPS , NMS , tardive movement disorders. Liver function abnormalities.	Additive effects with other CNS depressants, anticholinergics, alpha-adrenergic antagonists; inhibitors of cytochrome P450 enzymes (e.g., TCAs , fluoxetine, fluvoxamine, paroxetine) may increase serum levels; inducers of cytochrome P450 enzymes (e.g., carbamazepine, phenytoin) may decrease serum levels; effects of levodopa may be inhibited.	Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection. IM administration may be used to provide symptomatic control in patients with acute agitation or those who are unable to take the drug orally. Switch to the oral route as soon as practical, due to greater risk of EPS with IM administration. Not to be used in antipsychotic-naïve patients with first-episode psychosis. Intended for use in acute psychosis for up to 2 wk.
zuclopenthixol decanoate long-acting injection Clopixol Depot < \$25/200 mg vial	Range: 100–400 mg IM Q2–4 wk Usual: 150–300 mg IM Q2–4 wk	More common with low-potency agents: sedation, cardiovascular effects, anticholinergic effects, weight gain, lower seizure threshold, photosensitivity. More common with high-potency agents: increased prolactin, EPS , NMS , tardive movement disorders. Liver function abnormalities.	Additive effects with other CNS depressants, anticholinergics, alpha-adrenergic antagonists; inhibitors of cytochrome P450 enzymes (e.g., TCAs , fluoxetine, fluvoxamine, paroxetine) may increase serum levels; inducers of cytochrome P450 enzymes (e.g., carbamazepine, phenytoin) may decrease serum levels; effects of levodopa may be inhibited.	Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.

Drug/Cost ^[a]	Dosage ^[b]	Adverse Effects ^[c]	Drug Interactions	Comments
Drug Class: Antipsychotics, first-generation, high potency				
flupentixol Fluanxol \$25–50	Initial: 3 mg/day PO	More common with low-potency agents: sedation, cardiovascular effects, anticholinergic effects, weight gain, lower seizure threshold, photosensitivity. More common with high-potency agents: increased prolactin, EPS , NMS , tardive movement disorders. Liver function abnormalities.	Additive effects with other CNS depressants, anticholinergics, alpha-adrenergic antagonists; inhibitors of cytochrome P450 enzymes (e.g., TCAs , fluoxetine, fluvoxamine, paroxetine) may increase serum levels; inducers of cytochrome P450 enzymes (e.g., carbamazepine, phenytoin) may decrease serum levels; effects of levodopa may be inhibited.	Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.
	Usual: 3–6 mg/day PO			
	Maximum: 12 mg/day			
	Divided in 3 doses/day			
flupentixol decanoate long-acting injection Fluanxol Depot < \$25/20 mg vial	Range: 20–100 mg IM Q2–3 wk	More common with low-potency agents: sedation, cardiovascular effects, anticholinergic effects, weight gain, lower seizure threshold, photosensitivity. More common with high-potency agents: increased prolactin, EPS , NMS , tardive movement disorders. Liver function abnormalities.	Additive effects with other CNS depressants, anticholinergics, alpha-adrenergic antagonists; inhibitors of cytochrome P450 enzymes (e.g., TCAs , fluoxetine, fluvoxamine, paroxetine) may increase serum levels; inducers of cytochrome P450 enzymes (e.g., carbamazepine, phenytoin) may decrease serum levels; effects of levodopa may be inhibited.	Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.
	Usual: 20–40 mg IM Q2–3 wk			
fluphenazine generics < \$25	Initial: 2.5–10 mg/day divided in 3–4 doses PO	More common with low-potency agents: sedation, cardiovascular effects, anticholinergic effects, weight gain, lower seizure	Additive effects with other CNS depressants, anticholinergics, alpha-adrenergic antagonists; inhibitors of cytochrome P450 enzymes (e.g., TCAs , fluoxetine, fluvoxamine, paroxetine) may increase serum levels; inducers of	Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.
	Usual: 1–5 mg/day PO			
	Maximum: 20 mg/day			

Drug/Cost ^[a]	Dosage ^[b]	Adverse Effects ^[c]	Drug Interactions	Comments
haloperidol generics < \$25	Initial: 1.5–3 mg/day PO	threshold, photosensitivity. More common with high-potency agents: increased prolactin, EPS , NMS , tardive movement disorders. Liver function abnormalities.	cytochrome P450 enzymes (e.g., carbamazepine, phenytoin) may decrease serum levels; effects of levodopa may be inhibited.	
	Usual: 4–12 mg/day PO			
	Maximum: 20 mg/day			
	Divided in 1–3 doses/day			
haloperidol lactate immediate-acting injection generics < \$25	Usual range: 2–5 mg IM Q4–8H PRN	More common with low-potency agents: sedation, cardiovascular effects, anticholinergic effects, weight gain, lower seizure threshold, photosensitivity. More common with high-potency agents: increased prolactin, EPS , NMS , tardive movement disorders.	Additive effects with other CNS depressants, anticholinergics, alpha-adrenergic antagonists; inhibitors of cytochrome P450 enzymes (e.g., TCAs , fluoxetine, fluvoxamine, paroxetine) may increase serum levels; inducers of cytochrome P450 enzymes (e.g., carbamazepine, phenytoin) may decrease serum levels; effects of levodopa may be inhibited.	Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.
	Maximum: 20 mg/day			
		Liver function abnormalities.		IM administration may be used to provide symptomatic control in patients with acute agitation or those who are unable to take the drug orally. Switch to the oral route as soon as practical, due to

Drug/Cost ^[a]	Dosage ^[b]	Adverse Effects ^[c]	Drug Interactions	Comments
				greater risk of EPS with IM administration.
haloperidol decanoate long-acting injection generics	Range/usual: 50–300 mg IM Q4 wk Maximum: 450 mg Q4 wk	More common with low-potency agents: sedation, cardiovascular effects, anticholinergic effects, weight gain, lower seizure threshold, photosensitivity. More common with high-potency agents: increased prolactin, EPS , NMS , tardive movement disorders.	Additive effects with other CNS depressants, anticholinergics, alpha-adrenergic antagonists; inhibitors of cytochrome P450 enzymes (e.g., TCAs , fluoxetine, fluvoxamine, paroxetine) may increase serum levels; inducers of cytochrome P450 enzymes (e.g., carbamazepine, phenytoin) may decrease serum levels; effects of levodopa may be inhibited.	Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection. Dose is approximately 10–15 × daily oral dose.
\$75–100/100 mg vial		Liver function abnormalities.		
pimozide generics	Initial: 2–4 mg/day PO Usual: 2–12 mg/day PO Maximum: 20 mg/day; in poor metabolizers of CYP2D6 maximum: 4 mg/day Once daily dosing	More common with low-potency agents: sedation, cardiovascular effects, anticholinergic effects, weight gain, lower seizure threshold, photosensitivity. More common with high-potency agents: increased prolactin, EPS , NMS , tardive movement disorders. Liver function abnormalities. Pimozide: QT _c prolongation with doses >8 mg/day.	Additive effects with other CNS depressants, anticholinergics, alpha-adrenergic antagonists; inhibitors of cytochrome P450 enzymes (e.g., TCAs , fluoxetine, fluvoxamine, paroxetine) may increase serum levels; inducers of cytochrome P450 enzymes (e.g., carbamazepine, phenytoin) may decrease serum levels; effects of levodopa may be inhibited. Pimozide: avoid use with sertraline due to increased risk of QT _c prolongation.	Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.
< \$25				
trifluoperazine generics	Initial: 2–15 mg/day in 2–3 divided doses PO	More common with low-potency agents: sedation, cardiovascular	Additive effects with other CNS depressants, anticholinergics, alpha-adrenergic antagonists;	Advise patients about antipsychotic-associated body temperature
< \$25				

Drug/Cost ^[a]	Dosage ^[b]	Adverse Effects ^[c]	Drug Interactions	Comments
	Usual: 15–20 mg/day PO Maximum: 40 mg/day Divided in 1–2 doses/day	effects, anticholinergic effects, weight gain, lower seizure threshold, photosensitivity. More common with high-potency agents: increased prolactin, EPS , NMS , tardive movement disorders. Liver function abnormalities.	inhibitors of cytochrome P450 enzymes (e.g., TCAs , fluoxetine, fluvoxamine, paroxetine) may increase serum levels; inducers of cytochrome P450 enzymes (e.g., carbamazepine, phenytoin) may decrease serum levels; effects of levodopa may be inhibited.	dysregulation and prevention of heat stroke, e.g., hydration, sun protection.
^[a] Cost of 30-day supply of mean usual dose unless otherwise specified; includes drug cost only. ^[b] Majority of patients will respond to dosages at the lower end of the 'usual' dosage range. ^[c] For more information on management of adverse effects, see Table 5. Abbreviations: CNS = central nervous system; CV = cardiovascular; EPS = extrapyramidal side effects; NMS = neuroleptic malignant syndrome; TCA = tricyclic antidepressant				
Table 7: Drug Therapy for Psychosis: Second-Generation Antipsychotics				
Drug/Cost ^[a]	Dosage ^[b]	Adverse Effects ^[c]	Drug Interactions	Comments
Drug Class: Antipsychotics, second-generation				
aripiprazole Abilify, generics	Initial: 10–15 mg/day PO. Best taken in the morning due to risk of activating side effects (i.e., insomnia) In first-episode psychosis: 2–5 mg then gradually titrate to 10–15 mg/day Titration: If necessary, increase dose after 2 wk (due to long half-life) Maximum: 30 mg/day	EPS (akathisia, parkinsonism), dizziness, orthostatic hypotension, headache, GI complaints, nasopharyngitis, tremor, sedation, insomnia. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection. DRESS is a rare but very serious adverse effect.	Carbamazepine (or other strong inducers of CYP2D6 or CYP3A4, such as phenytoin, rifampin) can decrease aripiprazole levels significantly. Ketoconazole, quinidine, fluoxetine or paroxetine (or other strong inhibitors of CYP2D6 or CYP3A4) can increase levels substantially.	May also be classified as a third-generation antipsychotic. Partial agonist activity at D ₂ and 5HT _{1A} receptors and potent antagonism activity at 5HT _{2A} receptors (potential efficacy in treating negative and depressive symptoms).
\$150–350				
aripiprazole prolonged release	Ensure patient tolerates aripiprazole oral doses before initiating IM dose	EPS (akathisia, parkinsonism), dizziness, orthostatic hypotension, headache,	Carbamazepine (or other strong inducers of CYP2D6 or CYP3A4, such as phenytoin, rifampin) can	May also be classified as a third-generation antipsychotic. Partial agonist activity at D ₂ and

Drug/Cost ^[a]	Dosage ^[b]	Adverse Effects ^[c]	Drug Interactions	Comments
injectable suspension Abilify Maintena \$350–550	Initial: 400 mg IM × 1 dose; continue aripiprazole 10–20 mg daily PO × 14 days Alternative initial dose: 2 x 400 mg IM injections (at separate sites) plus 1 x 20 mg PO Maintenance: 400 mg monthly IM; if tolerability issues arise, may reduce to 300 mg monthly IM	GI complaints, nasopharyngitis, tremor, sedation, insomnia. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection. DRESS is a rare but very serious adverse effect.	decrease aripiprazole levels significantly. Ketoconazole, quinidine, fluoxetine or paroxetine (or other strong inhibitors of CYP2D6 or CYP3A4) can increase levels substantially. Avoid concomitant administration of CYP3A4 inducers for more than 14 days. Dose adjustments recommended when patients are adding or discontinuing concomitant strong inhibitors of CYP3A4 or CYP2D6.	5HT _{1A} receptors and potent antagonism activity at 5HT _{2A} receptors (potential efficacy in treating negative and depressive symptoms).

aripiprazole prolonged release injectable suspension Abilify Asimtufii \$1,000/dose	Ensure patient tolerates aripiprazole oral doses before initiating IM dose Initial: 960 mg IM × 1 dose; continue aripiprazole 10–20 mg daily PO × 14 days Alternative initiation strategies available in product monograph. Maintenance: 960 mg every 2 months IM; if tolerability issues arise, may reduce to 720 mg every 2 months IM	EPS (akathisia, parkinsonism), dizziness, orthostatic hypotension, headache, GI complaints, nasopharyngitis, tremor, sedation, insomnia. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection. DRESS is a rare but very serious adverse effect.	Carbamazepine (or other strong inducers of CYP2D6 or CYP3A4, such as phenytoin, rifampin) can decrease aripiprazole levels significantly. Ketoconazole, quinidine, fluoxetine or paroxetine (or other strong inhibitors of CYP2D6 or CYP3A4) can increase levels substantially. Avoid concomitant administration with strong CYP2D6 inhibitors and strong CYP3A4 inducers. Dose adjustments recommended when patients are adding or discontinuing strong inhibitors of CYP3A4.	May also be classified as a third-generation antipsychotic. Partial agonist activity at D ₂ and 5HT _{1A} receptors and potent antagonism activity at 5HT _{2A} receptors (potential efficacy in treating negative and depressive symptoms).
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asenapine Saphris < \$150	Initial: 5 mg BID SL Titration: target therapeutic dose	Oral hypoesthesia and paresthesia (occur after administration and usually resolve in 1 h), orthostatic hypotension,	May potentiate antihypertensive drug effects. Strong inhibitors of CYP1A2, such as	Derived from the antidepressant mirtazapine and is a potent multireceptor antagonist with strong affinity for a number of
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Drug/Cost ^[a]	Dosage ^[b]	Adverse Effects ^[c]	Drug Interactions	Comments
	same as initial dose (5 mg BID SL) Maximum: 10 mg BID SL	sedation, insomnia, EPS , constipation. Minimal effect on weight, glucose, lipids. Hypersensitivity reactions, often after first dose, have occurred. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.	fluvoxamine, increase asenapine levels. Other CYP1A2 inducers or inhibitors may affect asenapine levels.	serotonergic and dopaminergic receptors. ^[120] Instruct patients to handle the tablet with dry hands, to place it under the tongue immediately, to allow it to dissolve completely (do not chew or swallow) and to not drink or eat for 10 min afterward. If asenapine is taken in combination with other oral medications, it should be taken last. Inform patients of the signs and symptoms of a serious allergic reaction and advise them to seek immediate medical attention if one occurs.

brexpiprazole  Rexulti < \$150	Initial: 1 mg daily Titration: Day 1–4: 1 mg daily Day 5–7: 2 mg daily Depending on clinical response and tolerability, on Day 8, increase to 4 mg Reduce dose to maximum of 3 mg in patients with moderate-severe hepatic and renal impairment	Headache, insomnia or sedation, nausea, constipation, orthostatic hypotension, EPS . Cases of hyperglycemia, dyslipidemia, weight gain and sleep apnea have been reported. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.	Ketoconazole, quinidine, fluoxetine or paroxetine (and other strong inhibitors of CYP2D6 or CYP3A4) can increase levels substantially; modify dose as appropriate. Brexpiprazole levels can be reduced by strong CYP3A4 inducers (e.g., carbamazepine, phenobarbital, phenytoin, rifampin). Modify dose as appropriate. Not affected by smoking.	May also be classified as a third-generation antipsychotic. Partial agonist at the 5HT _{1A} , dopamine D ₂ and D ₃ receptors and as an antagonist at the 5-HT _{2A} , 5-HT _{2B} , 5-HT ₇ , noradrenergic alpha ⁻ _{1A} , alpha ⁻ _{1B} , alpha ⁻ _{1D} and alpha ⁻ _{2C} receptors. ^[40] This receptor profile leads to a favourable antipsychotic profile for the treatment of positive, negative, mood and cognitive symptoms of schizophrenia with a low risk of akathisia and EPS, weight gain and metabolic complications. ^[121]
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Drug/Cost ^[a]	Dosage ^[b]	Adverse Effects ^[c]	Drug Interactions	Comments
cariprazine Vraylar \$150–350	Initial: 1.5 mg daily Titration: increase gradually by 1.5 mg increments Usual: 1.5–6 mg once daily Maximum: 6 mg/day	EPS , akathisia, insomnia, restlessness, aggression, headache and hypertension. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.	Extensively metabolized by CYP3A4; contraindicated in combination with moderate/strong CYP3A4 inhibitors/inducers and for at least 2 wk after discontinuation (inhibitors).	Due to a long half-life, adverse effects may appear several weeks after initiation and dose adjustments. May also be classified as a third-generation antipsychotic. Partial agonist of D ₃ , D ₂ , 5-HT _{1A} . Higher affinity for D ₃ receptors may have potential to treat cognitive impairment and anhedonia.
clozapine (including oral disintegrating tablets) Clozaril, Gen-Clozapine, other generics \$150–350	Initial: 12.5–25 mg/day PO Titration: increase by 12.5–25 mg on second day and then by 25–50 mg daily PO depending on tolerance Usual: 300–600 mg/day PO Maximum: 900 mg/day Divided in 1–3 doses/day	Agranulocytosis (<1%), seizures (1–5%; dose-related), sedation, orthostatic hypotension, tachycardia, fever, nausea, weight gain, hypersalivation (30–50%), urinary incontinence, constipation; increased risk of diabetes and hyperlipidemia; myocarditis and other cardiac effects. Hypersalivation and sedation may contribute to pneumonia and subsequent toxicity. Constipation may evolve into potentially fatal intestinal obstruction. DRESS is a rare but very serious adverse effect. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.	Additive sedation with CNS depressants; may potentiate antihypertensive drug effects; inhibitors of CYP1A2, such as diltiazem, fluvoxamine or propranolol, or of CYP3A4, such as clarithromycin, erythromycin, grapefruit juice or prednisone, may increase clozapine levels; inducers of CYP1A2 or CYP3A4, such as carbamazepine, phenytoin, rifampin or cigarette smoking, may reduce clozapine levels; respiratory depression with higher doses of benzodiazepines; avoid use with bone marrow suppressants and drugs that lower the seizure threshold.	A few case reports describe elevated clozapine levels and toxicity after COVID-19 vaccination or natural infection, possibly due to inflammation-induced inhibition of CYP1A2. ^[122] Close monitoring after vaccination/infection is warranted. Associated with a greater risk of hematological malignancies (absolute risk low at 0.7%). ^[123]

Drug/Cost ^[a]	Dosage ^[b]	Adverse Effects ^[c]	Drug Interactions	Comments
lurasidone  Latuda, generics < \$150	Initial: 40 mg daily PO (with food) Titration: target therapeutic dose same as initial dose (40 mg daily PO) Usual: 40–80 mg daily PO Maximum: 160 mg daily PO; ^[124] consultation with a psychiatrist is recommended prior to increasing the dose above 80 mg daily	Nausea and vomiting, sedation, EPS , orthostatic hypotension, insomnia. ^{[40][125]} Nausea common at initiation and with dose increases and can be persistent. It is often worst 2–3 h postdose (time of peak plasma levels) and not related to presence or absence of food. DRESS is a rare but very serious adverse effect. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.	May potentiate antihypertensive drug effects; do not use in combination with strong inhibitors of CYP3A4, such as ketoconazole; do not exceed doses of 40 mg daily PO in combination with other inhibitors of CYP3A4, such as diltiazem, clarithromycin, erythromycin, prednisone, grapefruit juice (increased lurasidone levels); do not use in combination with strong inducers of CYP3A4, such as carbamazepine, phenytoin (decreased lurasidone levels).	Has high affinity for D ₂ , 5-HT _{2A} and 5-HT ₇ receptors, moderate affinity for alpha-adrenergic receptors, and little or none for muscarinic and histaminic receptors. Administer with meals of >350 kcal to maximize absorption and therapeutic effect. Nausea can be managed by lowering or splitting the dose (e.g., 40 mg twice daily rather than 80 mg once daily) or giving the dose at bedtime. Ginger or peppermint remedies may also be helpful.
olanzapine (including oral disintegrating tablets) Zyprexa, Zyprexa Zydis, generics < \$150	Initial: 5–10 mg/day PO Titration: increase by 2.5–5 mg every 3–4 days Usual: 10–20 mg/day Maximum: 20 mg/day (product monograph); doses of up to 40 mg/day are used in clinical practice under care of a psychiatrist Frequency: 1 dose/day; higher doses may be given in 2 divided doses	Weight gain, dizziness, sedation, anticholinergic effects, hepatic aminotransferase elevation, orthostatic hypotension, increased risk of diabetes and dyslipidemia, EPS (especially akathisia). DRESS is a rare but very serious adverse effect. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.	Sedation with CNS depressants; may potentiate antihypertensive drug effects; inhibitors of CYP1A2 or CYP2D6, such as diltiazem, fluvoxamine or paroxetine, may increase olanzapine levels; inducers of CYP1A2 or CYP3A4, such as barbiturates, carbamazepine, phenytoin, rifampin or cigarette smoking may decrease olanzapine levels.	Highest frequency of weight gain and metabolic abnormalities among SGAs.
olanzapine injection	Initial: 5–10 mg IM	Weight gain, dizziness, sedation, anticholinergic	Sedation with CNS depressants; may	Highest frequency of weight gain and metabolic

Drug/Cost ^[a]	Dosage ^[b]	Adverse Effects ^[c]	Drug Interactions	Comments
Zyprexa Intramuscular < \$150	Usual: 10 mg IM; if necessary, second dose of 5–10 mg IM may be given 2 h after first injection Maximum: 20 mg/day (oral and IM) with no more than 3 injections in 24 h	effects, hepatic aminotransferase elevation, orthostatic hypotension, increased risk of diabetes and dyslipidemia, EPS (especially akathisia). DRESS is a rare but very serious adverse effect. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.	potentiate antihypertensive drug effects; inhibitors of CYP1A2 or CYP2D6, such as diltiazem, fluvoxamine or paroxetine, may increase olanzapine levels; inducers of CYP1A2 or CYP3A4, such as barbiturates, carbamazepine, phenytoin, rifampin or cigarette smoking may decrease olanzapine levels. Should not be administered simultaneously with parenteral benzodiazepines due to reports of cardiac and respiratory problems including deaths.	abnormalities among SGAs.

paliperidone  Invega	Initial: 3–6 mg daily PO	Insomnia, headache, weight gain, orthostatic hypotension, rhinitis, anxiety, dose-related hyperprolactinemia, EPS .	Paliperidone is the active metabolite of risperidone. It is not metabolized by the liver and therefore has minimal risk of drug-drug interactions.	Less orthostatic hypotension and weight gain and more insomnia compared with risperidone.
\$150–350	Titration: if necessary, increase by 3 mg/day at 5-day intervals Maximum: 12 mg daily	Risk of intraoperative floppy iris syndrome in patients undergoing cataract surgery who have been exposed to paliperidone. DRESS is a rare but very serious adverse effect. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.	Carbamazepine may decrease paliperidone serum concentrations.	

paliperidone  palmitate long-acting injectable Invega Sustenna	Initial: 150 mg IM (deltoid muscle) and 100 mg IM (deltoid muscle) 7 days later; no oral	Insomnia, headache, weight gain, orthostatic hypotension, rhinitis, anxiety, dose-related hyperprolactinemia, EPS .	Paliperidone is the active metabolite of risperidone. It is not metabolized by the liver and therefore has minimal risk of drug-drug interactions.	Less orthostatic hypotension and weight gain and more insomnia compared with risperidone.
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Drug/Cost ^[a]	Dosage ^[b]	Adverse Effects ^[c]	Drug Interactions	Comments
\$350–550/dose	supplementation required Usual: 100 mg IM monthly in either deltoid or gluteal muscle Range: 25–150 mg IM monthly	Risk of intraoperative floppy iris syndrome in patients undergoing cataract surgery who have been exposed to paliperidone. DRESS is a rare but very serious adverse effect. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.	Carbamazepine may decrease paliperidone serum concentrations.	Patient acceptance and adherence may be improved compared to risperidone long-acting injections (Q2 wk).

paliperidone  palmitate long-acting injectable Invega Trinza	Initial/usual: 175–525 mg Q3 months IM Multiply Invega Sustenna dose by 3.5 to determine appropriate dosage	Insomnia, headache, weight gain, orthostatic hypotension, rhinitis, anxiety, dose-related hyperprolactinemia, EPS . Risk of intraoperative floppy iris syndrome in patients undergoing cataract surgery who have been exposed to paliperidone. DRESS is a rare but very serious adverse effect. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.	Paliperidone is the active metabolite of risperidone. It is not metabolized by the liver and therefore has minimal risk of drug-drug interactions. Carbamazepine may decrease paliperidone serum concentrations.	Less orthostatic hypotension and weight gain and more insomnia compared with risperidone. Must first establish Invega Sustenna as a tolerable and effective treatment for ≥4 months, with the last 2 doses equivalent in strength, before starting Invega Trinza. Patient acceptance and adherence may be improved compared to risperidone long-acting injections (Q2 wk) and other paliperidone long-acting injections (Sustenna; Q4 wk).
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Drug/Cost ^[a]	Dosage ^[b]	Adverse Effects ^[c]	Drug Interactions	Comments
quetiapine immediate-release Seroquel, generics < \$150	Initial: 50–100 mg/day PO	Sedation, dizziness, weight gain, orthostatic hypotension, hepatic aminotransferase elevation, headache, anticholinergic effects, increased risk of diabetes and dyslipidemia, possible increased risk of cataracts; may reduce thyroid hormone levels. DRESS is a rare but very serious adverse effect.	Additive sedation with CNS depressants; may potentiate antihypertensive drug effects; inhibitors of CYP3A4 (e.g., clarithromycin, erythromycin, grapefruit juice, ketoconazole, prednisone) may increase quetiapine levels; inducers of CYP3A4 (e.g., carbamazepine, phenytoin, rifampin) may decrease quetiapine levels.	Sedation and cognitive effects may be more prominent due to antihistaminic and anticholinergic properties.
	Titration: increase by 100 mg/day Usual: 600 mg/day Maximum: 800 mg/day (product monograph); doses of up to 1200 mg/day used in clinical practice under care of a psychiatrist Divided in 1–3 doses/day Although BID dosing usually recommended, many patients can be maintained on a single daily dose at bedtime ^[126]			
quetiapine extended-release Seroquel XR, generics < \$150	Initial: 300 mg QHS PO (200 mg for first-episode psychosis)	Sedation, dizziness, weight gain, orthostatic hypotension, hepatic aminotransferase elevation, headache, anticholinergic effects, increased risk of diabetes and dyslipidemia, possible increased risk of cataracts; may reduce thyroid hormone levels. DRESS is a rare but very serious adverse effect.	Additive sedation with CNS depressants; may potentiate antihypertensive drug effects; inhibitors of CYP3A4 (e.g., clarithromycin, erythromycin, grapefruit juice, ketoconazole, prednisone) may increase quetiapine levels; inducers of CYP3A4 (e.g., carbamazepine, phenytoin, rifampin) may decrease quetiapine levels.	Sedation and cognitive effects may be more prominent due to antihistaminic and anticholinergic properties.
	Titration (rapid): may increase dose in increments of ≤300 mg/day at intervals ≥1 day Usual: 400–800 mg/day Given as a once-daily dose, generally in the evening			
		Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.		

Drug/Cost ^[a]	Dosage ^[b]	Adverse Effects ^[c]	Drug Interactions	Comments
risperidone  Risperdal , Apo-Risperidone, other generics < \$150	Initial: 1–2 mg/day PO	Sedation, headache, weight gain, orthostatic hypotension, rhinitis, anxiety, dose-related hyperprolactinemia, EPS . Risk of intraoperative floppy iris syndrome in patients undergoing cataract surgery who have been exposed to risperidone.	Additive sedation with CNS depressants; may potentiate antihypertensive drug effects; inhibitors of CYP3A4 and CYP2D6 may increase risperidone levels; inducers of CYP3A4 may decrease risperidone levels. Dosage adjustments may be required.	Highest frequency of EPS among SGAs.
	Titration: increase by 0.5–2 mg PO every 3–7 days Usual: 4–6 mg/day PO Maximum: 10 mg/day and 8 mg/dose PO Frequency: 1 dose/day, preferably QHS			
		DRESS is a rare but very serious adverse effect. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.		
risperidone long-acting injection Risperdal Consta \$350–550	Initial: 25 mg IM every 2 wk (oral supplementation with current antipsychotic required for first 3 wk)	Sedation, headache, weight gain, orthostatic hypotension, rhinitis, anxiety, dose-related hyperprolactinemia, EPS . Risk of intraoperative floppy iris syndrome in patients undergoing cataract surgery who have been exposed to risperidone.	Additive sedation with CNS depressants; may potentiate antihypertensive drug effects; inhibitors of CYP3A4 and CYP2D6 may increase risperidone levels; inducers of CYP3A4 may decrease risperidone levels. Dosage adjustments may be required.	Highest frequency of EPS among SGAs.
	Titration: depending on response, increase by 12.5 mg every 4–8 wk Usual: 25–37.5 mg IM every 2 wk; some patients can be maintained on a dose of 12.5 mg every 2 wk Maximum: 50 mg IM every 2 wk			
		DRESS is a rare but very serious adverse effect. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.		

Drug/Cost ^[a]	Dosage ^[b]	Adverse Effects ^[c]	Drug Interactions	Comments
		Adverse effects may be less severe compared with oral risperidone due to decreased peak to trough serum fluctuations. Injection site pain is common.		
risperidone  long-acting injection Okedi	Ensure patient tolerates risperidone oral doses before initiating IM dose	Sedation, headache, weight gain, orthostatic hypotension, rhinitis, anxiety, dose-related hyperprolactinemia, EPS . Risk of intraoperative floppy iris syndrome in patients undergoing cataract surgery who have been exposed to risperidone.	Additive sedation with CNS depressants; may potentiate antihypertensive drug effects; inhibitors of CYP3A4 and CYP2D6 may increase risperidone levels; inducers of CYP3A4 may decrease risperidone levels. Dosage adjustments may be required.	Highest frequency of EPS among SGAs.
\$500–\$675/dose	Maintenance: 75–100 mg IM every 4 wk Equivalent to 3–4 mg/day PO; regimens outside of this range may not be eligible for this LAI	DRESS is a rare but very serious adverse effect. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection. Adverse effects may be less severe compared with oral risperidone due to decreased peak to trough serum fluctuations. Injection site pain is common.		
ziprasidone Zeldox, generics	Start with 40 mg BID PO (20 mg BID for antipsychotic-naïve first-episode patients) and rapidly titrate in the first wk up to 120–160 mg/day Recommended titration schedule for early psychosis	Orthostatic hypotension, EPS , sedation, insomnia. Prolonged QT _c -interval; contraindicated in patients with history of QT _c prolongation, recent MI, uncompensated heart failure or with concomitant use of any	Contraindicated with any other drug that prolongs the QT _c interval, such as antibiotics (azithromycin, clarithromycin, erythromycin), antiarrhythmics (amiodarone, ibutilide, procainamide, quinidine,	Has agonist activity at 5HT _{1A} receptors and, unlike other SGAs, has antagonist activity at 5HT _{1D} receptors and moderate inhibition of synaptic reuptake of serotonin and norepinephrine. Administer with meals of ≥500 kcal to maximize

Drug/Cost ^[a]	Dosage ^[b]	Adverse Effects ^[c]	Drug Interactions	Comments
	outpatient; ^[127] Day 1: 20 mg AM, 40 mg PM Day 2: 40 mg AM, 60 mg PM Day 3: 60 mg AM, 60 mg PM Day 4: reassess	other drug that prolongs the QT _c interval. Rapid titration is recommended to avoid ziprasidone-induced “activation” syndrome consisting of anxiety, restlessness, insomnia, increased energy and hypomanic-like symptoms, which develop soon after treatment initiation and occur at the lower end of the dosage range (20–40 mg BID PO).	sotalolol), lithium, methadone, venlafaxine. May potentiate antihypertensive drug effects.	absorption and therapeutic effect.
	Once-daily doses may be appropriate for some patients, but BID is typically required due to short half-life. Administer doses with food (see Comments)	DRESS is a rare but very serious adverse effect. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.		

^[a] Cost of 30-day supply of mean usual dose unless otherwise specified; includes drug cost only.

^[b] Majority of patients will respond to dosages at the lower end of the ‘usual’ dosage range.

^[c] For more information on incidence and management of adverse effects, see [Table 4](#) and [Table 5](#).

 Dosage adjustment may be required in renal impairment; see [Dosage Adjustment in Renal Impairment](#).

Abbreviations: CNS = central nervous system; DRESS = drug reaction with eosinophilia and systemic symptoms; EPS = extrapyramidal side effects; GI = gastrointestinal; SGA = second-generation antipsychotic

Suggested Readings

[Addington J, Addington D, Abidi S et al. Canadian treatment guidelines for individuals at clinical high risk of psychosis. *Can J Psychiatry* 2017;62\(9\):656–61.](#)

[American Psychiatric Association. *Practice guideline for the treatment of patients with schizophrenia*. 3th ed. Washington \(DC\): American Psychiatric Publishing; 2021.](#)

[Barnes TR, Drake R, Paton C et al. Evidence-based guidelines for the pharmacological treatment of schizophrenia: updated recommendations from the British Association for Psychopharmacology. *J Psychopharmacol* 2020;34\(1\):3–78.](#)

[Ricciardi L, Pringsheim T, Barnes TRE et L. Treatment recommendations for tardive dyskinesia *Can J Psychiatry* 2019;64\(6\):388–99.](#)

Schneider-Thoma J, Chalkou K, Dorries C et al. Comparative efficacy and tolerability of 32 oral and long-acting injectable antipsychotics for the maintenance treatment of adults with schizophrenia: a systematic review and network meta-analysis. *Lancet* 2022;399(10327):824-36.

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